# BES–$H_2O_2$–Ac

Hydrogen peroxide ( $H_2O_2$ ) detection probes.

> Manufacturer : Wako
> Molecular Formula : $C_{28}H_{11}F_7O_8S$
> M.W. = 640.44 g / mol

# Procedure

## Step 1: Prepare stock solution ( 10 mM )

Dissolve 1 mg powder in **156.2 μL DMSO** to form a 10 mM stock solution. Store at room temperature, must be **protected from light**.

## Step 2: Stain

### Short time incubation

Dilute to **10 - 50 μM** with growth medium. Incubate at room temperature for ≈ 20 mins, protected from light.

### Long time incubation

1. Make medium contains **100 nM** BES-$H_2O_2$-Ac.

- Add **4 μL** of 10 mM stock solution to **400 mL** medium before the medium solidified ( ~ 50°C ), mix thoroughly, must be protected from light.

2. Grow the plant on this medium for ≈ 6 days, cut the root and observe using confocal microscope.

## Step 3: Confocal parameters

| Parameters | Theoretical value | Our machine |
|---|---|---|
| Excitation peak | 485 nm | 488 nm |
| Emission peak | 515 nm | 505 ~ 541 nm |
| Magnification | | 5X |
| Laser intensity ( **488 nm** ) | | 20.0% |
| Master Gain ( **488 nm** ) | | 700 V |
| Master Gain ( **T-PMT** ) | | 240 V |
| Digital Gain | | 1.0 |
| Pinhole | | ≈ 35 $\mu$m |
| Z-stack interval | | 16 $\mu$m |
| Scan speed | | 6 |
| Scan direction | | ↔ |

## Step 4: Image processing

1. Open the CZI file with ImageJ, view stack with `Hyperstack`.

2. Generate BES and T-PMT layers

- Repeat this process twice to generate both the BES and T-PMT layers:
- `Image` ⟶ `Stacks` ⟶ `Z Project...` ⟶ Projection type: `Max intensity` ⟶ `File` ⟶ `Save As` ⟶ `Tiff...`

3. Generate stacks

- Load in the previously generated BES and T-PMT TIFF files.
- `Image` ⟶ `Stacks` ⟶ `Images to Stack`

4. Measure distance

- Measure the distance between root tip and the BES signal distal boundary. The root tip can be easily recognized in the T-PMT layer.

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Done !!!

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